Team: #6

A Satellite-Optimized Multimodal Corridor to Bypass the Hormuz Crisis

Team members: William Blair, Sonali Bhalla, Jay Rao, Anushka Raghavendra

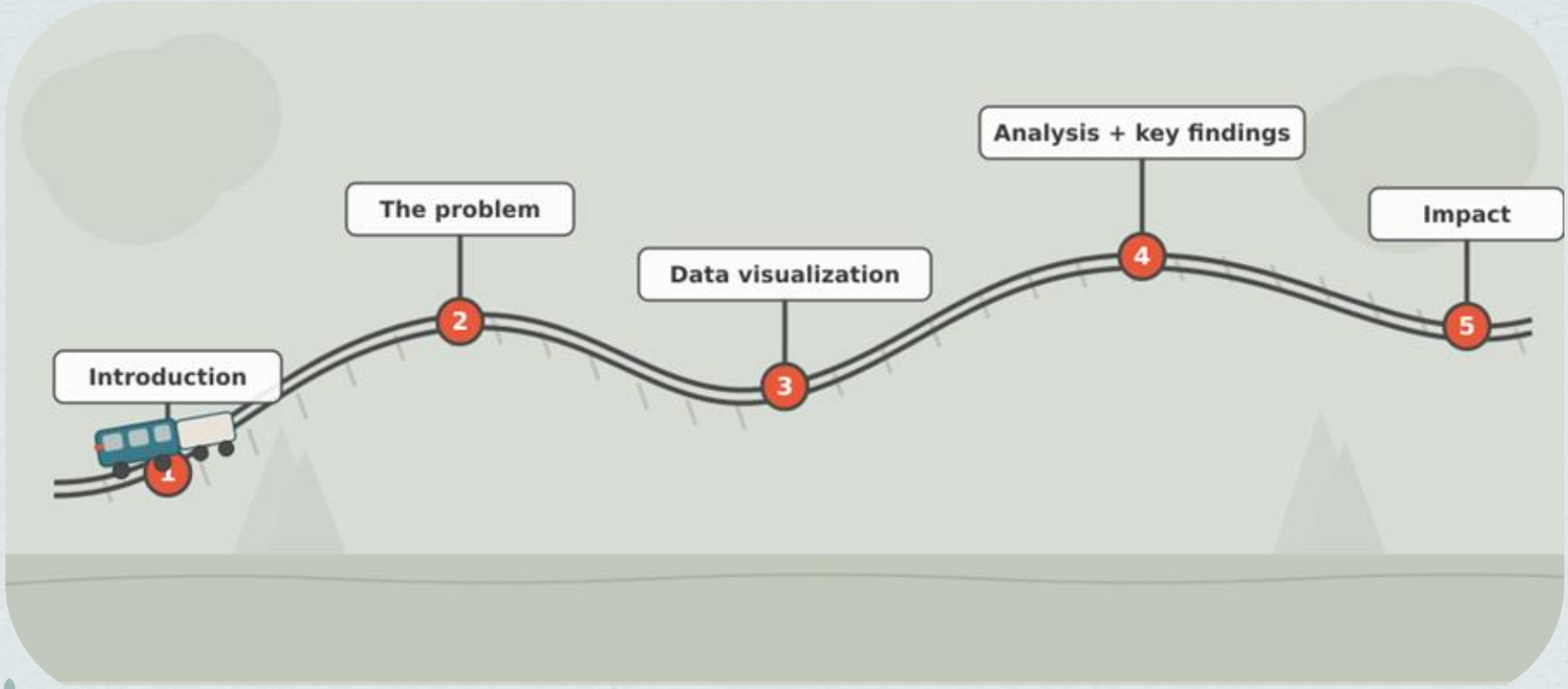
Track: Innovating the Net-Zero Supply Chain

Institution: Arizona State University



SpaceHACK for Sustainability

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Team Members



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Name: Jay Rao
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Major: Electrical Engineering
Role on Team: Research Expert

Approach – The Problem We Investigated

The February 28, 2026 Hormuz closure forced ~200 ships/week to reroute around Africa, adding ~77% more CO_2 per voyage.



The Solution

A satellite-optimized electric freight railway from Jubail to Haifa, powered by trackside solar arrays.



Contextual Factors & Stakeholders

Socio-Economic

- Rail stimulates economic growth and can make cities more accessible for supply chain
- Big investment for building & creating lines, but due to its longevity, it ends up creating positive financial impact

Political

- Govt policy to incentivize project being built
  - Ex: construction subsidies
- Mandatory reporting of sustainability initiatives

Cultural

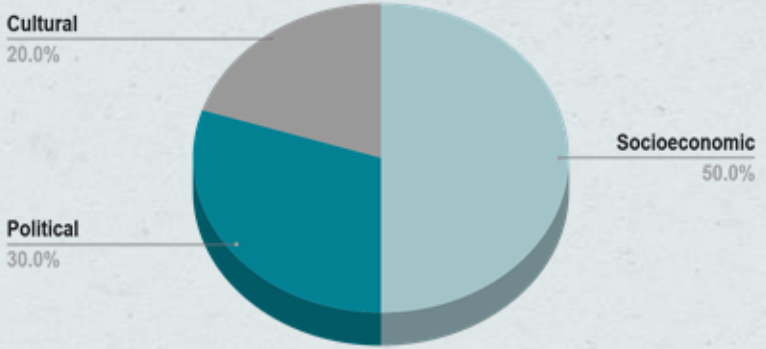
- Community engagement and trust

Key Stakeholders

- Government agencies, rail operators, communities, landowners, financiers, businesses

Relevant Policies

- Land-use and zoning laws
- Environmental regulations
- Cross-border trade and customs agreements



Connecting EO data to the track goal

Track goal: To achieve net-zero emission transportation of goods by optimizing routing, reducing time, and minimizing fuel consumption

Topography

Identifying optimal routes with lower fuel consumption.

CO

Emission of CO has drastically increased compared to 2025

Elevation

To support efficient rail route planning

Satellite based vessel tracking

To analyze real-time ship movement and identify congestion in shipping routes



Executive Summary

Initial Results

- Land bridge with electric rail saves ~**1.47M** tonnes CO_2 / week versus Cape reroute
- Solar electric rail achieves **96%** emissions reduction over diesel trucks on the overland segment
- Annual savings potential: **2.5 million** tonnes CO_2 (equivalent to **540,000** cars off the road)

Opportunities for further exploration

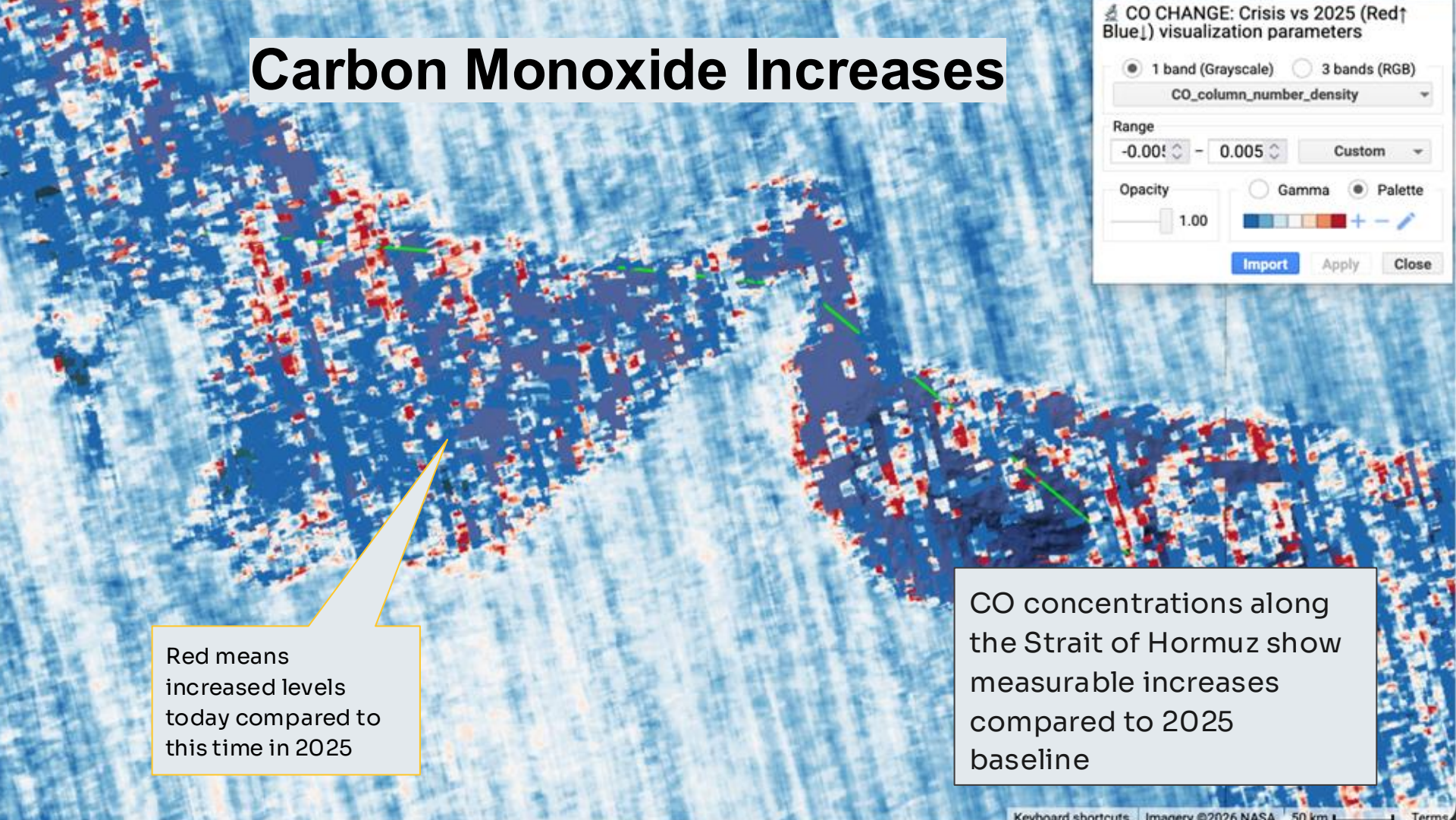
- Detailed engineering study of the optimized low-elevation corridor (your SRTM route avoids the Ha'il plateau)
- Cost-benefit analysis of trackside solar array installation along the **1,700** km route
- Integration with the planned GCC **Unified Railway Network**
- **Real-time routing optimization platform** using live Sentinel-5P and VIIRS data

Lessons Learned

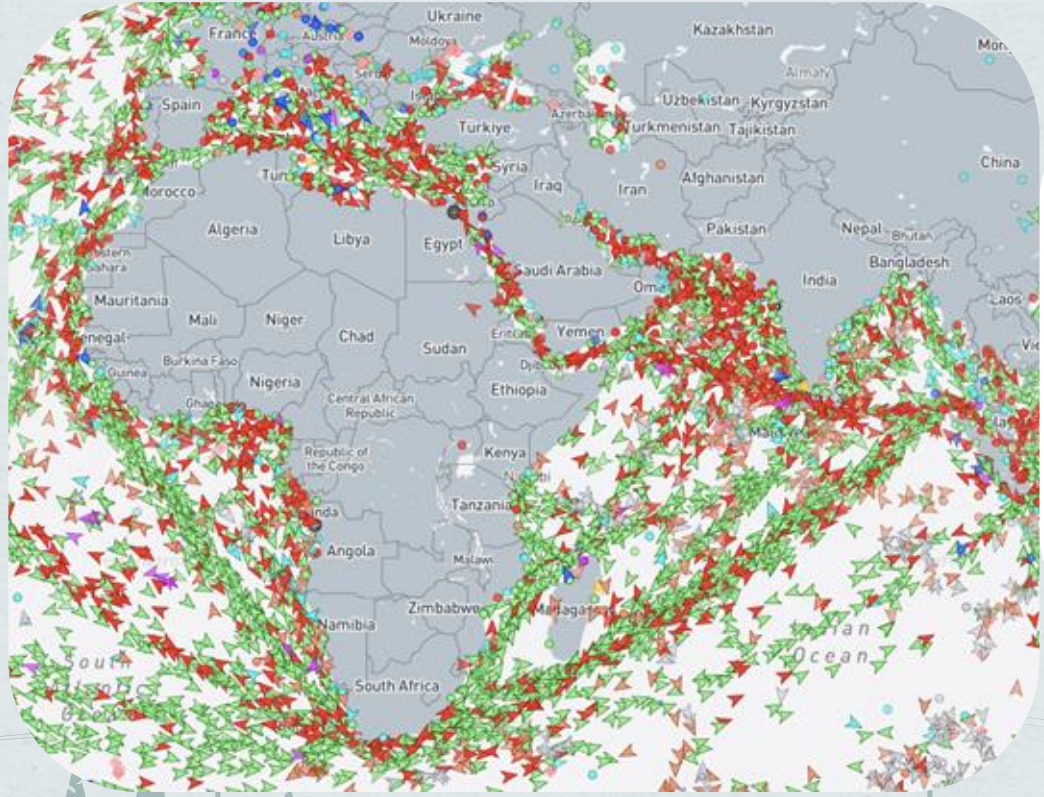
- High upfront costs, but due to **long-term use** it has **positive financial impact**
- Ship emissions over open ocean produce a weak satellite signal — port-area measurements were more reliable
- Initially explored aircraft as a modality but eliminated it due to emissions per TEU being far worse than any surface mod

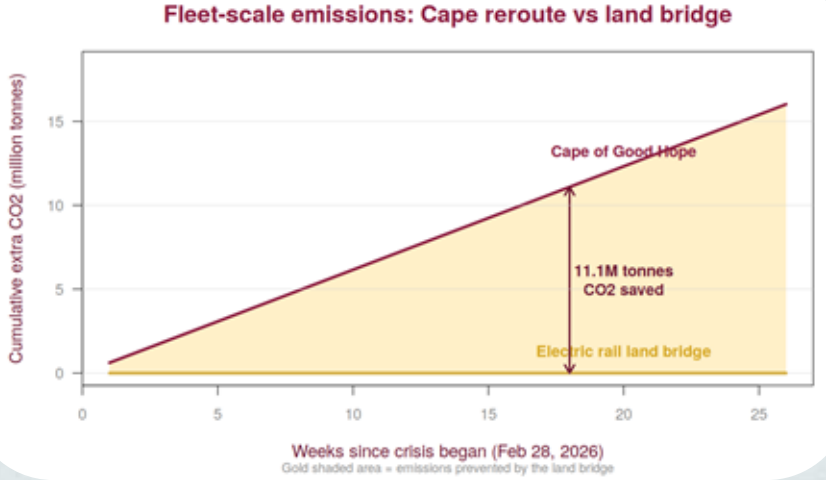


Carbon Monoxide Increases



Cape reroute vs Land bridge

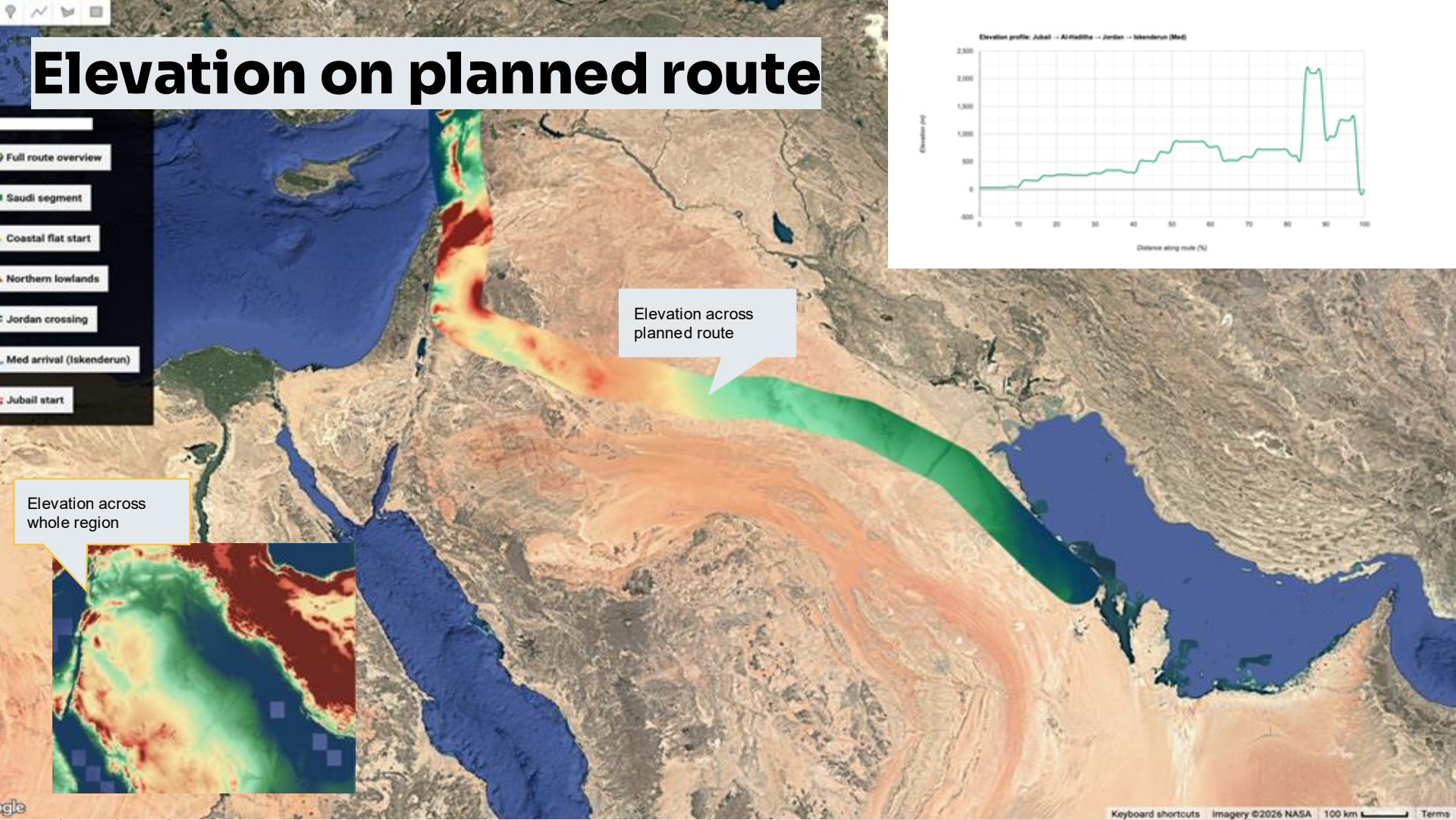




Proposed Rail Route



Elevation on planned route



Satellite Datasets



Every claim is grounded in satellite data

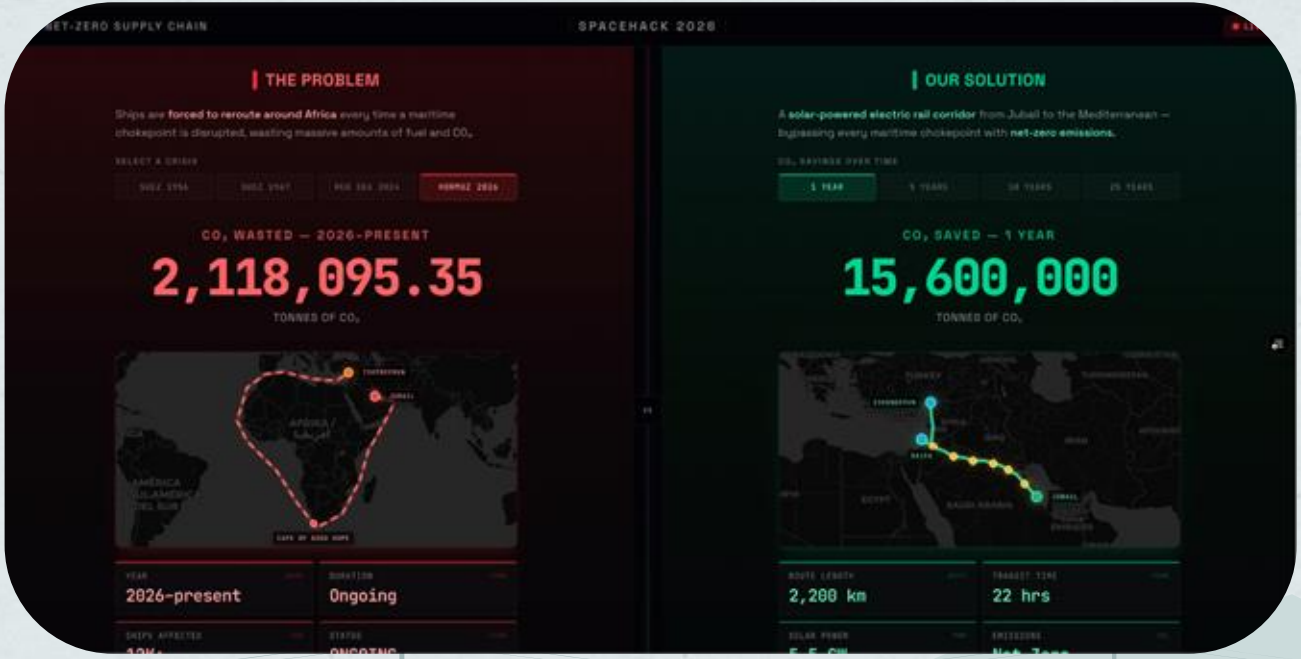
Google Earth Engine datasets → decisions

Sentinel-5P NO_2/SO_2	Pollution pathways shift when fleets detour — Cape corridor lights up vs Hormuz *Problem evidence (emissions & flows)*
Sentinel-5P CO	Carbon monoxide traces combustion intensity — validates higher fuel burn on Cape detour *Emissions verification (combustion footprint)*
SRTM Elevation	Terrain-aware routing — lowest-energy overland alignment for rail + PV *Engineering optimization*
ERA-5 Solar/Temp	High confidence on solar resource + thermal stress on diesel alternatives *Net-zero enabler (renewables + climate)*



Click this link for live demo

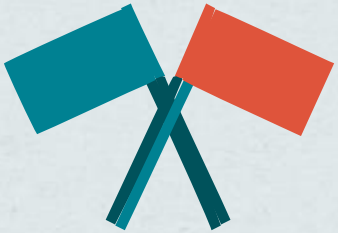
Net-Zero Supply Chain Corridor



Key Findings

Problem Statement

“How can the maritime, port, trucking, and aviation sectors collaborate to achieve true decarbonization and create sustainable “green corridors” for global trade?”



Key Questions

1. How do electric trains reduce the amount of emission? How feasible is it to implement?
2. What supply chain transportation methods release the least amount of carbon emissions?
3. How can we optimize supply chain logistics to increase efficiency (esp during times of crisis)?
4. Can we use multi-modal transportation in order to increase route optimization?
5. How can geospatial data help measure decarbonization?

Key Findings

1. **Find most efficient route** using satellite data including terrain, elevation, & CO2 levels
2. The Strait of Hormuz causes **major congestion** for ships, **increasing CO_2** emissions
3. Rail **improves efficiency** for bulk cargo
4. Creating a rail system costs lots of money and emissions, but long term it has positive financial and environmental impact.

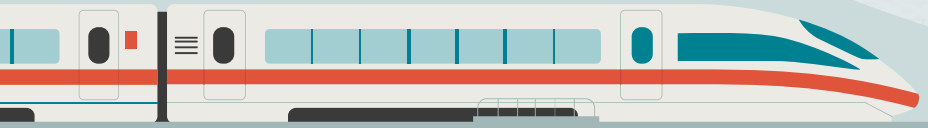
Why Earth Observation (EO) Data Matters

- EO data revealed a **hidden emissions spike** caused by rerouting around Africa after the **Hormuz disruption**
- CO satellite data shows **significant increases** compared to 2025, confirming real environment impact
- **Topographic analysis** for route optimization by minimizing elevation changes
- Satellite data helps optimize for efficiency and **reduce our carbon footprint.** We can place the **solar fields** in the best position and also reduces distance and elevation change to improve efficiency.



11.1 Million tonnes of CO_2 saved

over the first 18 weeks of crisis if the land bridge had been operational



$1.5–$2M

saved per vessel in avoided fuel costs per Cape diversion, multiplied across hundreds of ships per month



Impact/Why

- Saudi Arabia Railways activated this exact freight corridor on March 26, 2026 — validating the concept in real time
    - We are using electric trains in the efforts to create reduced CO_2 emissions
- High upfront costs, but due to long-term use it has **positive financial** and **environmental impact**
- Green Corridors created due to rail implementation as it is both sustainable and has potential to create seamless transportation of goods
- Freight rail produced approx **26g CO_2 per net tonne km**
    - **3-4 times more efficient than road freight**

96%
Emissions reduced per TEU*

85%
Fewer emissions used than planes

76%
Fewer emissions used than cars



*by switching from diesel trucks (1.904t CO_2) to solar-electric rail (0.085t CO_2)



Thank You!

Icon pack



Sources

1. OcoonScore (2024). Bunker consumption analysis: Cape of Good Hope rerouting. Industry report. 2. IMO (2023). Fourth Greenhouse Gas Study. International Maritime Organization, London. 3. Sea-Intelligence / BIMCO (2024). Container shipping rerouting cost and emissions impact report. 4. Vomer et al. (2024). CO emissions from container shipping rerouting via the Cape of Good Hope. Environmental Science & Technology. 5. UNCTAD (2024). Review of Maritime Transport. United Nations, Geneva. 6. IEA (2023). Rail Handbook — Energy Consumption and CO Emissions. International Energy Agency, Paris. 7. European Environment Agency (2023). Freight transport emissions: comparison by mode. EEA Technical Report. 8. IPCC (2022). AR6 Working Group III — Transport chapter. IPCC Sixth Assessment Report. 9. IRENA (2024). Renewable energy cost curves — Gulf Cooperation Council region. Abu Dhabi. 10. EIA (2024). Strait of Hormuz: World's most important oil chokepoint. U.S. Energy Information Administration. 11. S&P; Global Commodity Insights (2026, March). Hormuz closure impact assessment. Market report. 12. Kpler (2026, February–March). Tanker transit data and carrier suspension tracker. Real-time data. 13. Bloomberg Markets (2026, February). Oil price spike analysis — Hormuz closure Feb 28, 2026. 14. Saudi Railways Organisation (SRO). North-South Railway project data and cost benchmarks. Riyadh. 15. GCC Secretariat (2009, updated 2022). GCC Unified Railway Network — feasibility and cost studies. 16. McKinsey Global Infrastructure Initiative (2023). Infrastructure cost benchmarks: Middle East. 17. CEIC Data (2024). Infrastructure construction cost database — Middle East and North Africa. 18. Google Earth Engine. 19. SolarTech. 20. EY Oxford College of Procurement & Supply. World Bank. OECD.

